TED (15)3093 Reg No…………..

(RIVISION – 2015) Signature……….

THIRD SEMESTER DIPLOMA MODEL EXAMINATION IN POLYMER TECHNOLOGY

SYNTHETIC RUBBER Time: 3Hrs

(Maximum Marks: 100)

PART - A

(Maximum Marks: 10)

I Answer all questions in one or two sentences. Each question carries 2 marks.

1. Identify any four General purpose rubbers?
2. Write the monomer with structure of EPM?
3. Write the excellent property and any two applications of polysulphide rubbers?
4. Expand HTPB and XNBR?
5. Define blending? (5X2=10)

PART - B

(Maximum Marks: 30)

II Answer any five of the following questions. Each question carries 6 marks.

1. Explain the monomer with structure and preparation of the monomer of SBR?
2. What is the purpose of adding Diene monomers in EPDM? What are they?
3. CR is considered as an outstanding special purpose rubber. Why?
4. Explain the polymerization, properties and applications of polyurethane?
5. Write a short note on Ionomers?
6. Describe the production and applications of thermoplastic S-I-S?
7. What are the advantages of polymer blends? (5X6=30)

PART -C

(Maximum Marks: 60)

(Answer one full question from each MODULE. Each full question carries 15 marks)

MODULE -I

III	a.	Explain the polymerization and applications of BR?	7
	b.	Explain the manufacture, properties and applications of IIR?	8

OR

IV	a.	Explain the manufacture, properties and applications of SBR?	8
	b.	Explain the manufacture, properties and applications of IR?	7

MODULE –II

V	a.	Explain the monomers with structure, polymerization and applications of NBR?	8
	b.	Explain the polymerization, properties & applications of Polysulphide rubbers?	7

OR

VI	a.	Explain the polymerization, properties & applications of Silicone rubber?	8
	b.	Explain the structure, polymerization & applications of CSM?	7

MODULE -III

VII	a.	Explain the polymerization, properties & applications of Carboxylated SBR?	8
	b.	Explain the production, properties & applications of Hydrogel?	7

OR

VIII	a.	Explain the polymerization, properties & applications of Acrylic rubbers?	8
	b.	Explain the polymerization, properties and applications of fluorocarbon rubbers?	7

MODULE - IV

IX	a.	Explain TPE?	8
	B.	Explain the production, properties & applications of thermoplastic EVA?	7

OR

X	a.	Explain the production, properties & applications of thermoplastic Polyester?	7
	b.	Explain blending procedure, properties and application of NBR/PVC blend?	8